Municipal air-quality decision-support pipeline

Batyrova, Yerkezhan | Matriculation no. 92131651 | DLBDSEDE02 Project: Data Engineering | Portfolio part 3 - Abstract

Problem framing and stakeholder requirements

A municipal air-quality pipeline is useful only when its outputs correspond to real decisions. The redesigned prototype therefore starts with five roles. Environmental analysts need raw values, missingness and anomaly flags to judge whether a trend is credible. City planners and policy teams need station comparisons and before/after date windows, including observation counts, to discuss transport or land-use interventions. Public-dashboard developers need a stable latest-reading and daily-series contract. Data stewards need run status, heartbeats, rejected rows and reconciliation counters. Authorised warning officers may review threshold candidates, but public notification is outside this prototype and must remain blocked until a municipality approves thresholds, ownership and escalation rules.

These needs become measurable acceptance criteria. A load must reconcile all source rows as stored or quarantined, create no duplicate station-hour keys, survive a replay without changing the stored count, expose terminal run status and quality totals, and support four representative queries within a configurable 2,000 ms target in the declared local environment. The latency target is a prototype service objective, not a universal performance claim; the included benchmark stores elapsed time and fails its command when any query misses the target.

Source, bias and fitness for use

The UCI Beijing Multi-Site Air Quality dataset contains 420,768 hourly records from 12 stations for 2013-2017 (UCI Machine Learning Repository, n.d.). It is used as a historical test fixture because its volume, multiple stations and missing fields exercise the ingestion design. It is not the intended production feed and cannot justify conclusions about another city. The supplied CSV data alone do not establish sensor calibration, precision, maintenance history, station coordinates, siting quality or independence between nearby stations. A government source is therefore recorded as provenance, not treated as proof of accuracy. Hourly values may be adequate for operational trends yet inappropriate for short exposure events or policy indicators requiring other averaging periods (World Health Organization, 2021).

The pipeline makes those limits visible. Missing measurements remain absent instead of becoming zero. Technical plausibility ranges are configurable and explicitly separated from legal or health limits. Questionable but structurally usable observations remain queryable with warning flags; malformed rows enter a quarantine collection with a reason. Aggregations return observation counts, allowing users to see whether comparisons rely on unequal evidence. Production onboarding would add a station registry with coordinates, calibration and sensor model, signed delivery manifests, independent-reference checks and a domain-approved aggregation policy.

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Implementation and maintainability

The Python loader downloads the official archive, recognises its nested ZIP structure and streams the twelve station CSV files. Each row becomes a timezone-aware station-hour document. Deterministic identifiers and insert-only upserts make replays harmless; batches of 1,000 use bounded retry for transient errors. The flexible measurements object accepts future sensor fields, while JSON Schema validation protects identity, timestamp, provenance and quality fields (MongoDB, n.d.-a).

MongoDB was chosen for indexed station/time access and evolving measurement sets. PostgreSQL suits a stable relational registry and join-heavy analysis; files lack concurrent indexed queries; a warehouse adds premature scope. Separate collections hold measurements, quarantine, runs, benchmarks and source metadata. Docker Compose packages the prototype, while production still requires secrets, TLS, backups, replication and workload-led sharding.

Evaluation and learning

Eleven unit tests cover transformation, missing values, future metrics, batching, nested archives, quarantine IDs, quality flags and query contracts. A complete parse read all 420,768 rows across 12 stations: 383,964 passed every configured check and 36,804 remained available with warnings. In 36,788 rows a required pollutant was missing; 16 contained an out-of-range value; categories can overlap. No row was structurally rejected. The repository includes the reproducible profile, health and quality commands, and latency benchmark. Docker was unavailable in the document-generation environment, so infrastructure latency is not claimed; the executable benchmark is the target-machine acceptance gate.

Feedback showed that naming users is insufficient: each needs a decision, data contract, quality evidence and failure condition. This changed a generic loader into an auditable service boundary. Next, municipal interviews should validate needs; a registry should capture stations and calibration; spatial coverage, averaging and alert policies need domain approval; and concurrent workloads require benchmarking before a dashboard consumes the contracts.

References

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